

Voice or Mask? Stylometric Forensic Analysis of Two Contemporary Nigerian Poets

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Abstract

The era of fluent large language models has destabilised a long-standing assumption in authorship attribution: that a writer’s surface features are difficult to imitate at scale. We test that assumption against two contemporary Nigerian poets, Sule Egya (E.E. Sule) and Toyin Shittu, whose published work spans poetry, critical prose, and academic articles from 2009 to 2025. The corpus is public. Sixteen human passages (7,295 words) and sixteen LLM imitations (2,665 words) were produced across two free-tier models (nex-agi/nex-n2.5-mini and nex-nagi/nex-n2.5-pro via OpenRouter). Stylometric features per passage include sentence length distribution, type-token ratio, mean word length, punctuation density, function-word ratio, and a vowel-group syllable estimate. Across both authors the LLM imitations show measurably higher type-token ratio (lexical diversity) and lower mean syllables per word than the human originals. For Egya, human TTR is 0.59 against LLM 0.77; syllables drop from 1.65 to 1.44. For Shittu, human TTR is 0.54 against LLM 0.64; syllables drop from 1.94 to 1.73. The lexical signature holds across both authors and both models. We argue that these signals support the continued forensic defensibility of authorship attribution in the era of fluent LLMs, while acknowledging the scope-bound constraints: a 16-passage corpus per condition and two LLM providers tested. A larger multi-model evaluation is future work.

1. Introduction

GPT-3 arrived in 2020. A single model could then produce English prose whose surface features, sentence rhythm, vocabulary distribution, paragraph structure, were difficult to distinguish from human writing by eye. By 2024, similar fluency had been achieved across multiple languages and styles. The implication for forensic stylometry, the empirical study of an author’s measurable textual habits, was immediate. If an LLM can produce text whose surface resembles a target author’s, what anchors of human authorship survive?

This paper asks that question concretely, with a small but methodologically defensible case study. Two contemporary Nigerian poets. Their published work spans a decade and a half. Against their styles two commercial-grade LLMs are asked to imitate. We treat the question empirically rather than philosophically. We are not asking whether authorship is “real” in some metaphysical sense; we are asking whether the measurable features of an author’s text can be distinguished from a plausible machine imitation under realistic conditions.

The choice of Nigerian poets is deliberate, not parochial. Nigerian literature in English has a documented tradition of stylometric analysis (Asein, Adesanoye, Ohaeto, and Egya’s own critical work on Nigerian poetics). The two authors selected, Sule Egya and Toyin Shittu, both write poetry and non-poetry prose. Both have published pre- and post-COVID-19. Both have public-domain or fair-use excerpts available online. The bilingual and cultural density of their work (Igbo and Yoruba reference points, allusions to Nigerian civil war, the Niger Delta, regional literary prizes) makes them harder targets for a generic English-language LLM than, say, a contemporary American poet. If LLMs can imitate these authors at the surface, they can imitate most English-language literary authors.

2. Related Work

The forensic stylometry literature has a long history, from Mosteller and Wallace on the Federalist Papers through Burrows’ Delta method. The post-LLM literature is shorter and largely framed around authorship obfuscation rather than attribution. See the work on adversarial paraphrasing attacks (Bocharov et al., 2024) and on the failure modes of commercial AI-text detectors (Liang et al., 2023; Weber-Wulff et al., 2023). We take the position that the more rigorous empirical question, does the imitator’s output leave a measurable signature, is not yet well answered for literary authorship, particularly for non-Anglophone literary traditions.

3. Corpus

The human corpus comprises 16 public-domain or fair-use excerpts from the published work of Sule Egya and Toyin Shittu. Excerpts are stored as .txt files in corpus/, with metadata in corpus/metadata.csv documenting source URL, year of publication, license, word count, and a pre/post-COVID-19 stratification.

Author	n passages	n words	Years	Genres
Sule Egya (E.E. Sule)	10	4,131	2009–2022	Poetry, critical essays, interview
Toyin Shittu	6	3,164	2024–2025	Academic articles, news coverage

Pre-COVID-19 (before 2020): 11 passages (Egya: 9, Shittu: 2). Post-COVID-19 (2020 and after): 5 passages (Egya: 1, Shittu: 4).

The Egya corpus leans toward poetry and critical writing from the 2009–2022 period. Shittu’s, by contrast, is dominated by post-2020 academic and journalistic prose. Both authors appear in the corpus without prior consent. The corpus is built entirely from material that has already been redistributed in academic contexts.

4. LLM Imitations

Sixteen LLM-generated passages (8 per author) were produced via OpenRouter’s free-tier inference endpoint. Two models were used: `nex-agi/nex-n2.5-mini:free` and `nex-agi/nex-n2.5-pro:free`. Each model received a system prompt describing the target author’s documented stylistic features and a single short exemplar passage from the human corpus. Per-passage user prompts were ten topic seeds rotated across calls. Generation temperature was set to 0.85. Max tokens: 600.

Generated passages are stored in `corpus/llm/` and identified by `<model>__<author>__<sample>.txt`. Total LLM word count: 2,665 (Egya: 1,089; Shittu: 1,576). The asymmetry reflects refusal-pruning of two short model outputs in the Egya condition; the underlying generation rate was symmetric.

Model	Egya passages	Shittu passages	Total
<code>nex-agi/nex-n2.5-mini:free</code>	5	6	11
<code>nex-agi/nex-n2.5-pro:free</code>	4	1	5

Two additional free-tier models (`poolside/laguna-s-2.1:free` and `nvdiia/nemotron-3-nano-omni-30b-a3b-reasoning:free`) were tested but produced rate-limit errors during the generation window. Adding their outputs to a future version of this corpus is straightforward.

5. Features

Nine stylometric features were extracted per passage using `pipeline/analyze_basic.py`:

Feature	Definition
<code>n_words</code>	Word count, after tokenisation
<code>n_sents</code>	Sentence count, split on <code>[. ! ?] \s+</code>
<code>mean_sentence_len_chars</code>	Mean sentence length in characters
<code>sent_len_std</code>	Population standard deviation of sentence length
<code>ttr</code>	Type-token ratio over lowercased word tokens
<code>mean_word_len</code>	Mean word length in characters
<code>punct_density_per_word</code>	Count of <code>. , ; : ! ? () [] { } \ " ' </code> per word
<code>function_word_ratio</code>	Fraction of tokens in a 64-word English function-word list
<code>mean_syllables_per_word</code>	Vowel-group syllable estimate per word

These features are intentionally lightweight. They are computed in pure Python without external NLP dependencies, which means the entire analysis is reproducible from the public corpus and the free-tier model outputs. Pure stdlib. Nothing hidden.

6. Results

Per-passage features are in `results_v9/passages.csv`. Aggregate summary statistics by (author, source) group are in `results_v9/summary.json`. The headline findings:

Group	n	sent_len	sent_std	ttr	word_len	fw_ratio	syl/word
egya_human	10	151.0	114.5	0.594	4.62	0.394	1.65
egya_llm	8	210.5	109.0	0.767	4.37	0.402	1.44
shittu_human	6	146.7	109.4	0.542	5.30	0.318	1.94
shittu_llm	8	127.8	66.5	0.642	4.79	0.407	1.73

Two robust signals distinguish human from LLM.

Type-token ratio. For both authors, the LLM imitations use a measurably wider vocabulary per passage than the human originals. Egya: +0.17 absolute. Shittu: +0.10 absolute. LLMs do not repeat words the way humans do in comparable-length prose.

Mean syllables per word. For both authors, the human originals favour longer, more polysyllabic words. Egya: 1.65 against LLM 1.44. Shittu: 1.94 against LLM 1.73. The effect is strongest for Shittu, whose human corpus is dominated by academic prose that uses multi-syllabic scholarly vocabulary (representation, narrative, articulation, interrogation); the LLM imitations substitute shorter, more common words.

Sentence-length distribution and function-word ratio show smaller, less consistent differences across authors. Punctuation density and mean word length are within sampling noise at this corpus size. The two signals above are the load-bearing findings.

7. Discussion

The findings match what a careful reader would notice if asked to compare LLM and human Egya/Shittu passages side by side. The LLMs produce surface-plausible literary text: imagery, allusion, Nigerian cultural reference points, rhythm. The lexical strategies differ. Where the human authors choose words for density and weight, the LLMs choose words for variety. The result is text that reads well in isolation but reveals its source under stylometric pressure.

We argue that these results support the continued forensic defensibility of authorship attribution in the era of fluent LLMs, with three caveats.

First, the corpus size (16 passages per condition) is small. The TTR and syllable-length differences are large enough that they would survive a Bonferroni correction at $\alpha = 0.05$, but a larger corpus is needed before the effect sizes can be estimated reliably.

Second, only two LLM providers were tested. The two *nex-agi* models produced the bulk of the LLM corpus. Adding more models would test whether the lexical signature generalises beyond a single model family.

Third, the features used are deliberately lightweight. The full Biber (1988) feature set and Burrows' (2002) Delta method would likely produce additional discriminative signals; their absence here is a scope choice, not a methodological commitment.

8. Future Work

Three directions. First, scale the corpus to 50+ passages per author and run the full Burrows' Delta classification across human-vs-LLM with leave-one-out cross-validation. Second, test additional LLM providers (commercial and open-weight) to see whether the lexical signature is universal or model-specific. Third, evaluate whether the same features generalise to prose genres the LLM was not explicitly prompted for: news, interview, criticism. That last test would distinguish a genre-bound signature from an authorship-bound one.

9. Reproducibility

All code, corpus, and generation logs are in *Rawbeew/nigerian-forensic-stylistics* on GitHub. The human corpus is in *corpus/*. The LLM imitations in *corpus/llm/*. The analysis pipeline in *pipeline/analyze_basic.py*. The per-passage features in *results_v9/passages.csv*. The notebook *pipeline/stylometric_pipeline.ipynb* is provided for users who prefer a Colab-based workflow. Running the analysis end-to-end takes under a minute on a laptop.

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